AI-Powered Smart Campus Resource Management System

Design Project – I (E1CSA313 / 5CS1990) · Review–II Technical Defense Report

Alliance University · Alliance School of Advanced Computing (ASAC) | Department of Computer Science & Engineering

Batch ID: DP1-24-28-CSE-GEN-L-01 (Section L) | **Academic Year:** 2026–2027

Faculty Mentor: Dr. Rashmi Mothkur (Associate Professor, ASAC CSE) | **Course Coordinator:** Dr. M. Selvam

Live Production URL: https://smart-campus-digital-twin-eight.vercel.app

Team Members & Subsystem Ownership

Student Name	Registration No	Subsystem Role	Key Technical Deliverable
Likith S (Lead)	2411021061437	Graphics & Systems Architect	React 19, 3D WebGL (Three.js), Headless Blender pipeline, Vercel CI/CD
Vinodkumar	2411021061425	Microgrid Analytics Lead	D3 dynamic Sankey power visualizer, 1.2 MWh BESS peak shaving
Nikhil Kumar V	2411021061427	Data Ingestion Lead	Baseline ingestion (campusData.ts), TimescaleDB schema contracts
Kalyan Kumar T	2411021061421	Space Intelligence Lead	Constraint satisfaction timetable solver, 15-min hysteresis buffer

1. Requirement Analysis

- **Institutional Problem Formulation:** 28.4%–35.2% electrical energy is wasted through continuous HVAC operation in vacant classrooms across the 60-acre Anekal campus (Chintan Block peak draw: 14,200 kWh/day). Manual room bookings create timetable overlaps and high ghost-booking rates, leading to BESCOM peak maximum demand tariff penalties.
- **Functional Requirements:** Synchronize ERP academic timetables, detect ghost bookings, automatically resolve room allocation clashes via Constraint Satisfaction, balance campus microgrid power (grid + solar + battery), and provide centralized 3D spatial monitoring.
- **Engineering Scope Boundaries:**
  - *Strictly In-Scope (Software-First):* WebGL 3D digital twin (sustained 60 FPS), D3 Sankey vector flow router, timetable clash solver, 15-minute hysteresis buffer, calibrated historical BESCOM and solar baseline ingestion.
  - *Strictly Out-of-Scope (No New Physical Hardware):* Zero new physical sensors or microcontrollers installed on campus (compliant with ASAC Phase-1 software guidelines); zero 11 kV physical substation rewiring; zero biometric or camera surveillance (DPDP Act 2023 compliant).

- **Performance Benchmarks:** Sustained 60 FPS in standard browsers, 3D asset package <15 MB, sub-100ms state updates, 0 hard timetable clashes.

2. Module Description

- **Module 1 — 3D WebGL Digital Twin & Central Multi-Role Dashboard (Likith S):** React 19 + Three.js (r170) rendering the 60-acre campus topology generated via a headless Blender 4.3 Python script (`perfect_alliance_campus.py`). Features interactive building inspection drawers (occupancy, temperature, lux, W/m²), astronomical sun-angle lighting (06:00–22:00 IST), and auto-hiding dock actuators.
- **Module 2 — Microgrid Analytics & Dynamic D3 Sankey Power Visualizer (Vinodkumar):** Dynamic vector Sankey power visualizer balancing 11 kV BESCOM grid draw, 480 kWp rooftop solar PV, and 1.2 MWh BESS battery storage. Executes afternoon peak shaving to cut peak commercial grid demand by up to 18.4%.
- **Module 3 — Baseline Data Ingestion & Time-Series Pipeline (Nikhil Kumar V):** Software data pipeline operating without deploying new physical sensors. Ingests empirical institutional utility records (historical BESCOM bills: 380–420 kW chiller baseline load) and Bengaluru astronomical solar curves (12.845° N, 77.684° E) via standardized JSON schema contracts.
- **Module 4 — Space Intelligence & Timetable Constraint Solver (Kalyan Kumar T):** Google OR-Tools CSP room allocation solver achieving 0 hard clashes. Cross-references ERP schedules to detect ghost bookings and applies a strict **15-minute hysteresis buffer** to prevent false alarms during transient student breaks.

3. Convergence, Coverage and Completeness towards the Idea

- **Review–1 Feedback Closure:**
  1. *Dynamic Microgrid Visualization:* Engineered and deployed the dynamic D3 SVG Sankey visualizer showing live power transfers across grid, solar, and battery.
  2. *False Alert Suppression:* Integrated the 15-minute hysteresis sliding window in `campusEvolutionAgent.ts` to prevent premature chiller setbacks.
- **Full Campus Coverage:** Mapped complete 60-acre Anekal campus topology (Academic blocks, Library, Admin, Hostels, Sports, and Rooftop Solar).
- **System Convergence:** Unifies three isolated silos (ERP academic scheduling, spatial occupancy, and electrical microgrid operations) into a single operational interface.
- **Resolving the 8% PRISMA Research Gap:** Literature review of 148 smart campus systems revealed fewer than 8% worldwide integrate 3D spatial visualization with renewable microgrid dispatch and space intelligence.

4. Implementation (Exceeding 55%)

- **Milestone Verification:** Approved Semester V implementation scope stands at **>55% completed** (exceeding the 50% threshold for Review–II).
- **Live Production Build:** Publicly accessible at smart-campus-digital-twin-eight.vercel.app.

- **Live Verified Capabilities:** In-browser 60 FPS 3D campus navigation, building inspection drawer telemetry, real-time D3 Sankey energy flow, timetable room conflict resolution, and manual facility actuators.

5. Paper & Scientific Publication Status

- **Manuscript Title:** *"AI-Powered Smart Campus Digital Twin: Cyber-Physical Microgrid Energy Optimization and Space Intelligence"*
- **Status:** Completed 6-page IEEE two-column conference draft incorporating empirical benchmarks (60 FPS rendering, 18.4% peak shaving, 4.2 ghost hours recovered/week).
- **References & Plagiarism:** 14 peer-reviewed sources synthesized (`references.bib`); DrillBit similarity verified at **<8%** (compliant with <10% ceiling). Target: Scopus/IEEE-indexed conference track per mentor guidance.

6. Comprehensive Report and Presentation Deliverables

- **Formal Engineering Drawings Booklet:** 5-sheet architectural drawings (Sheet 1 Architecture, Sheet 2 DFDs, Sheet 3 UML & Hysteresis, Sheet 4 Microgrid SLD, Sheet 5 Master Site Plan).
- **Review–II Defense Deck:** 14-slide structured defense presentation (`02_Review_2_Presentation.pptx` / `.pdf`).
- **IEEE Term Paper Report:** 40+ page comprehensive technical document.
- **Review Diary:** Bound logbook with verified weekly entries through Weeks 10–12 prepared for signature.